Journal of Chromatography A 1713 (2024) 464565

Contents lists available at ScienceDirect

Journal of Chromatography A

journal homepage: www.elsevier.com/locate/chroma

Detailed comparison of in-house developed and commercially available heart-cutting and selective comprehensive two-dimensional liquid chromatography systems

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ARTICLE INFO

Keywords:
2D-LC
Mobile phase incompatibility
Heartcutting 2D-LC (LC-LC)
Selective comprehensive 2D-LC (sLC x LC)

ABSTRACT

Recently, two-dimensional liquid chromatography (2D-LC) has become a popular approach to analyze complex samples. This is partly due to the introduction of commercial 2D-LC systems. In the past, 2D-LC was carried out on in-house developed setups, typically consisting of several switching valves and sample loops as the interface between the two dimensions. Commercial systems usually offer different 2D-LC modes in combination with specialized software to operate the instrument and analyze the data. This makes them highly user-friendly, however, at an increased cost compared to in-house developed setups. This study aims to make a comparison between an in-house developed 2D-LC setup and a commercially available 2D-LC instrument. The comparison is made based on experimental differences, in addition to more general differences, including cost price, flexibility, and ease of operation. Special attention is also paid to the different strategies to deal with the mobile phase incompatibility between the highly orthogonal separation mechanisms considered in this work: hydrophilic interaction liquid chromatography (HILIC) and reversed-phase LC (RPLC). For the commercial 2D-LC instrument, this is done using active solvent modulation (ASM), a valve-based approach allowing the on-line dilution of the effluent eluting from the first dimension column before transfer to the second dimension (^{2}D) column. For the in-house developed setup, a combination of restriction capillaries and a trap column is used. Using a sample of 28 compounds with a large polarity range, peak shapes and recoveries of the ^{2}D-chromatograms are compared for both setups. For early eluting compounds, the selective comprehensive approach, currently only possible on the commercial 2D-LC instrument, results in the best peak shapes and recoveries, however, at the cost of an increased analysis time. In general, depending on the analytical goal (single heart-cut versus full-comprehensive 2D-LC), an in-house developed system can be satisfactory for the analysis of specific target compounds/samples. For more complex problems, it can be interesting to use a more specialized commercial 2D-LC instrument. Overall, this comparison study provides advice for analytical scientists, who are considering to use 2D-LC, on the type of equipment to consider, depending on the needs of their particular applications.

1. Introduction

Complex samples, containing a large number of compounds with diverse physicochemical properties, are increasingly encountered in various research fields, including environmental [1,2], food [3,4], (bio-) pharmaceutical analysis [5], proteomics [6,7], metabolomics [8,9] and lipidomics [10–12]. The characterization of these complex samples remains extremely challenging, due to the high separation power required for their adequate analysis [13].

To improve the separation power in liquid chromatography (LC), several strategies have been proposed, including the use of small particle packed columns operated at ultra-high pressure (UHPLC) [14], core-shell particles [15], monolithic stationary phases [16] and improvements in LC-mass spectrometry (LC-MS) instruments [17]. However, despite the impressive improvement in separation efficiency and peak capacity these developments offer, they are often still inadequate for very complex samples [18]. In such cases, two-dimensional liquid chromatography (2D-LC) can provide a solution. In 2D-LC, (part of or

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https://doi.org/10.1016/j.chroma.2023.464565
Received 21 October 2023; Received in revised form 21 November 2023; Accepted 5 December 2023
Available online 6 December 2023

the entire) effluent of a first separation dimension (^{1}D) column, typically containing unresolved peaks, is sent to a second-dimension (^{2}D) column for further separation [18]. When the separation mechanisms in both dimensions are sufficiently different or *orthogonal*, this results in an increased peak capacity [8,13,19]. Online 2D-LC approaches, wherein the ^{1}D-effluent is automatically sent to the ^{2}D using one or multiple switching valves equipped with sampling loops, are mostly used, because of the better reproducibility, reduced sample contamination and faster operation. On the other hand, offline approaches wherein the ^{1}D-effluent is temporarily collected offline before reinjection into the ^{2}D-column are typically less complex than online approaches, however, at the cost of being more time-consuming, less repeatable, and with an increased risk of sample contamination [20].

There are two main modes of 2D-LC: full comprehensive (LC x LC) and heart-cutting (LC-LC) 2D-LC. In LC × LC, which is usually the method of choice for highly complex samples, the entire ^{1}D-effluent is sampled and sent to a second column for additional separation in a ^{2}D-analysis time that equals the sampling time in ^{1}D. In LC-LC, only one fraction of a specific region of interest in the ^{1}D is analyzed in the ^{2}D, making this a useful method when additional separation is needed for selected compounds only, such as impurities coeluting with the active pharmaceutical ingredient in pharmaceutical analysis. Variations of these two modes are also possible, including multiple heart-cutting 2D-LC (mLC-LC), where fractions are taken of multiple regions of the ^{1}D-separation, and selective comprehensive 2D-LC (sLC × LC), which samples specific regions of the ^{1}D-separation in a comprehensive manner [21]. The key advantage of sLC × LC is its ability to break the link between the timescales of the ^{1}D- and ^{2}D-separations, allowing to address practical challenges related to under-sampling and solvent incompatibility. This advantage shows similarities to those derived from the offline 2D-LC approach, without the major drawbacks [22].

To achieve the highest possible resolving power for a 2D-LC method, the combined LC modes need to be complementary. This can be obtained by selecting different stationary and mobile phases in both dimensions. An important problem encountered when combining different separation mechanisms is their typical mobile phase incompatibility [23,24]. An example is the combination of hydrophilic interaction liquid chromatography (HILIC) and reversed-phase LC (RPLC), a useful combination for the analysis of samples containing compounds with a broad variety in polarity, as encountered in environmental, pharmaceutical and natural product analysis, metabolomics and proteomics [25–30]. The challenge for this specific combination is the opposite elution strength of their mobile phases [29]. In HILIC, the mobile phase typically contains a high percentage of acetonitrile (ACN), which is a strong eluent in RPLC. Together with the effect of a much larger injection volume compared to one-dimensional LC (1D-LC), the transfer of large ACN-rich fractions from ^{1}D-HILIC can negatively impact the peak shapes obtained in ^{2}D-RPLC [23,26,31].

If HILIC is used in the second dimension, the same problem arises, now caused by the large water-rich fractions transferred from ^{1}D-RPLC. This may result in band broadening, peak distortion, and/or analyte breakthrough, affecting the ^{2}D-separation by decreasing both the peak capacity and peak intensity [23,26]. Different strategies to deal with this mobile phase incompatibility have recently been explored [32–34]. In general, these include strategies to modify the composition and/or the volume of the fractions transferred from the first to the second dimension using either flow splitting [35–38], in-line mixing modulation (ILMM) [39], on-line dilution (with trapping columns [26,40,41] or sample loops/capillaries [29,42-44]), and solvent evaporation [45,46].

Despite these complications, the essential concepts of LC-LC and LC x LC were already introduced in the 1970s [10,47]. However, the number of publications involving 2D-LC only started to increase in the 1990s [48], and have seen a surge in the past decade [5]. An important factor in this increasing popularity is the transition from in-house developed 2D-LC instruments, originally used to investigate and improve the technology, to commercially available 2D-LC systems, offering a wide range of applications in a relatively straightforward way, making this technique accessible to less experienced users [21].

Commercially available 2D-LC systems can typically be used in various configurations, including single LC-LC, multiple LC-LC (mLC-LC), sLC x LC, and LC x LC, due to the presence of valves containing multiple sample loops. To deal with the mobile-phase incompatibility, different solutions are available. One example, called active solvent modulation (ASM), is a valve-based approach using a bypass capillary to dilute the ^{1}D-effluent with a weak solvent (compared to the ^{2}D-column) prior to transfer to the ^{2}D-column. The dilution solvent is provided by the same pump as used for the ^{2}D-separation, making this a relatively simple approach to implement in a 2D-LC setup. The dilution factor (DF) results from the split ratio between the ASM bypass capillary and the valve containing the sample loops ('deck'), and depends on the backpressures generated by the flow path through the ASM bypass capillary and the flow path through the deck (Fig. 1). The DF can be changed by changing the dimensions of the ASM bypass capillary [49].

Despite the increasing popularity of these commercial 2D-LC solutions, 2D-LC separations can also be performed on simpler, in-house modified 1D-LC systems. As an example, our research group developed an LC-LC setup that combines HILIC in ^{1}D with RPLC in ^{2}D, allowing to analyze polar and non-polar compounds within a single chromatographic run, using a single quaternary pump [26]. In this approach, HILIC and RPLC columns are combined using high-pressure switching valves that allow redirecting unretained non-polar compounds eluting from the HILIC column to a sample loop. To address the mobile phase incompatibility between both separation modes, an innovative mixing unit is used to dilute the ^{1}D-effluent in the sample loop before performing the ^{2}D-separation. This dilution process is based on a similar mechanism as the ASM approach. The mixing unit is assembled by interconnecting two restriction capillaries of different dimensions using three high-pressure switching valves and two T-pieces (Fig. 1). The sample loop is connected in series with a restriction capillary characterized by a high backpressure, which, in turn, is interconnected in parallel with a restriction capillary with a significantly lower backpressure. By applying a highly aqueous mobile phase through both restriction capillaries, the highly organic mobile phase containing the non-polar compounds in the sample loop is diluted to achieve a mobile phase composition suitable for the subsequent RPLC analysis. To ensure this dilution can be achieved at a sufficiently high flow rate, the non-polar compounds are first sent to a short RPLC trap column with a low backpressure after dilution. Subsequently, the trap column is connected to the RPLC column for analysis. This setup was shown to be capable of separating a mixture of common pharmaceuticals with a wide range in polarity (Log D values between −5.75 and 4.22) with satisfactory recoveries (between 82 % and 99 %) and repeatability [26]. Since only one pump is used for the entire operation, the separation can moreover be visualized in a single chromatogram, hence no specialized 2D-LC software is required. However, the setup in its current form only allows single heart-cutting 2D-LC analysis.

This study aims to investigate for the first time the capabilities of the in-house developed 2D-LC setup in-depth and compare it with a commercial 2D-LC system for a more complex sample of 28 pharmaceuticals and pesticides with a wide polarity range, as typically encountered in environmental analysis. Special attention is paid to the optimization of the different modulation mechanisms of both systems, especially regarding the different strategies to deal with the mobile phase incompatibility between HILIC and RPLC. On-line dilution on the commercial 2D-LC system is obtained via ASM and compared to the approach using restriction capillaries and a trap column in the in-house developed setup. For the commercial instrument, both LC-LC and sLC x LC are considered, while the in-house developed setup is only operated in LC-LC. An alternative approach to deal with the mobile phase incompatibility, called in-line mixing modulation (ILMM), is also considered for the commercial instrument [39]. ILLM is achieved by installing a commercially available in-line mixer (Jet Weaver, JW) in front of the

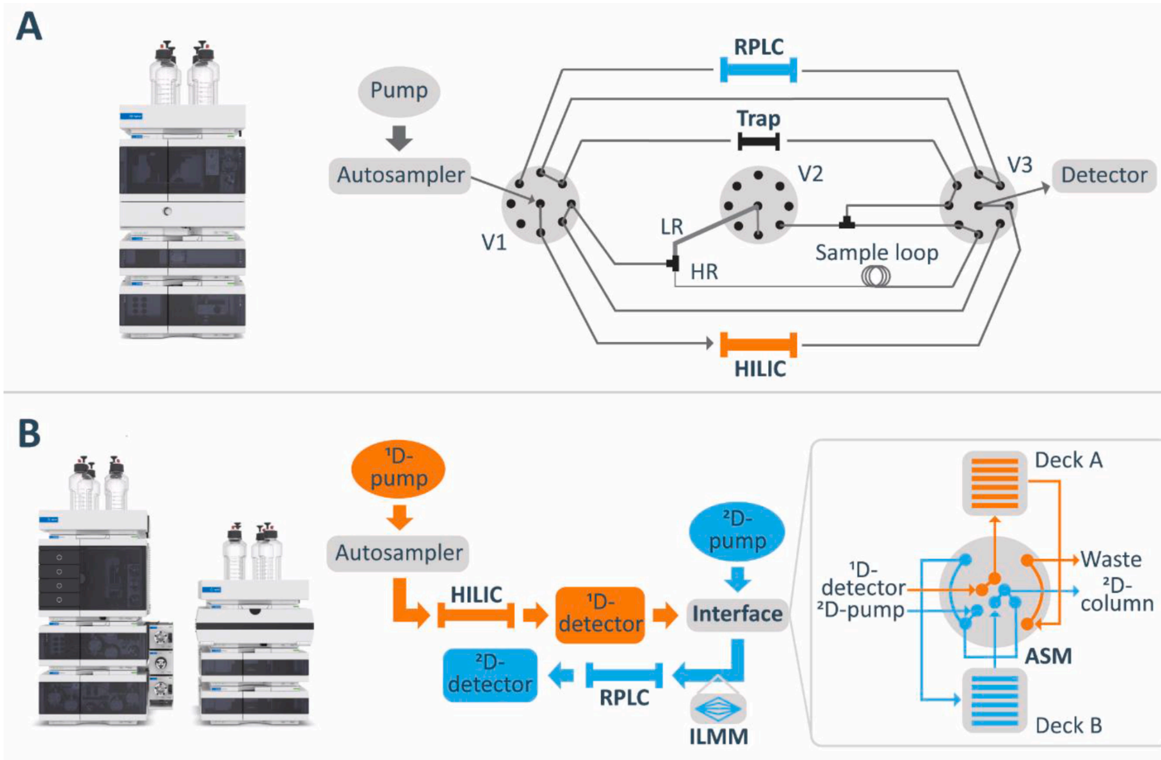

Fig. 1. Overview of the in-house developed (**A**) (V: valve, HR/LR: high/low restriction capillary) and the commercial (**B**) 2D-LC setup (ASM: Active Solvent Modulation, ILLM: In-line mixing modulation). (Images taken from Agilent Technologies [53,54]).

^{2}D-column (Fig. 1). JWs are normally installed after the LC pump head and used to homogenize mobile phases with different strengths and viscosities. They contain a three-dimensional multichannel flow path that distributes the flow at its inlet into multiple partial flows and recombines them again at the outlet. In ILLM, the DF can be changed by using a combination of JWs of various volumes [26,39]. A recent publication compared the performance of ILMM (using JWs with volumes of 35, 100 and 380 μL) to ASM (DF of 5) in various 2D-LC setups. In general, for larger ^{1}D-fractions, ASM was not sufficient to deal with the mobile phase incompatibility in each setup, while ILMM did improve the ^{2}D-peak shapes when larger JW mixer volumes were used for larger fractions, even in a higher solvent incompatibility scenario (higher transfer volume) in the LC x LC mode [39].

The two set-ups considered in this work are evaluated based on obtained peak shapes, peak widths and recoveries, as these are important parameters for an accurate quantification of compounds of interest in real samples, in addition to differences in analysis time. A more general comparison is also made based on the cost of the systems, their flexibility, ease of implementation, and robustness. Since both set-ups use components from the same vendor, any bias in terms of performance and cost price is moreover excluded.

2. Experimental

2.1. Chemicals

Caffeine, progesterone, pipamperone, bezafibrate, atenolol, metolachlor, (4)-nonylphenol, and ketoprofen were obtained from Sigma-Aldrich (Diegem, Belgium), sulfamethoxazole from Roche (Mannheim, Germany), ibuprofen and lidocaine from Certa (Braine-l'Alleud, Belgium), lorazepam from Fagron (Waregem, Belgium), amitrole from J&K Scientific (Marbach, Germany), and propranolol, paracetamol, diclofenac, diazepam, and carbamazepine from Alpha Pharma (Zwevegem, Belgium). Paroxetine, trimethoprim, iopromide, venlafaxine, codeine, cilostazol, imipramine, ticlopidine, oxazepam, and escitalopram were available in the lab as USP reference standards. The solvents used to prepare sample stock solutions and mobile phases were: acetonitrile (ACN, HPLC and MS grade) from Sigma-Aldrich and VWR (Leuven, Belgium), water (H_2O, MS grade) from VWR, ethanol (EtOH, absolute, analytical reagent grade) from Fisher Chemical (Loughborough, UK), and methanol (MeOH, HPLC grade) from Thermo Fisher Scientific. Formic acid (FA, 99 % for LC-MS) was from VWR and ammonium formate (AF, ≥ 99.995 %) was from Sigma-Aldrich. Ultrapure H_2O (conductivity = 0.1 μS/cm, pH 6.00) was produced in the lab using a Milli-Q gradient purification system from Millipore (Bedford, MA, USA).

2.2. Sample preparation

The sample evaluated in this study contained 28 compounds with a wide variety in polarity, as typically encountered in environmental analysis. All compounds, their sample solvents, concentrations in stock and working solutions and predicted log D values at pH 3.0 (calculated using Marvin Sketch software version 23.10, ChemAxon (https://www.chemaxon.com)) are shown in **Table S1** in the Supporting Information (SI). Sample stock solutions of all compounds were prepared in concentrations between 2 and 20 mg/mL, depending on the solubility of the compounds. Working solutions were made by mixing appropriate volumes of the different compounds and adding H_2O and ACN to obtain a total sample volume of 5 mL with compound concentrations of 10–400 μg/mL in 10:90 H_2O:ACN (v/v%), to ensure enough sample volume for all experiments on both 2D-LC instruments. These concentrations were selected to acquire compound peaks with comparable heights at a detection wavelength of 254 nm. For the recovery determination, an additional sample was prepared containing only the 15 non-polar compounds (compounds 14–28 in **Table S1**), with the same concentrations and solvent composition as in the above mixture. All samples were stored in the fridge (4 °C).

2.3. Instrumentation

A schematic overview of both systems is shown in Fig. 1.

2.3.1. In-house developed 2D-LC setup

The in-house developed 2D-LC setup consisted of an Infinity 1290 system from Agilent Technologies (Waldbronn, Germany) equipped with a 1290 quaternary pump, a 1290 autosampler with a maximum injection volume of 20 μL and a 1290 DAD detector with a 1.0 μL flow cell and a path length of 10 mm. The system was additionally equipped with three 9-port/8-position ultra-high-pressure valves (valves 1, 2 and 3, max. pressure of 1200 bar) from Agilent Technologies. NanoViper tubings (inner diameters ranging from 50 to 150 μm) from Thermo Scientific (Germering, Germany) or stainless steel tubing (diameter: 170 μm) from Agilent Technologies with the shortest possible length were used to establish connections between the pump, valves, columns and detector. The dwell volume of the quaternary pump was 550 μL. To collect the non-polar compounds eluting from the HILIC column, Rheodyne sample loops with volumes of 100, 50, and 33 μL, along with VICI Valco T-pieces featuring a bore size of 0.25 mm, were acquired from Sigma-Aldrich. Sample loops were connected in series with zero-dead volume unions to obtain the desired volume. Agilent MassHunter Workstation (version 10.0, Agilent Technologies) was used to operate the system, control the valve configurations and acquire and analyze data.

2.3.2. Commercial 2D-LC setup

The commercial 2D-LC system was a 1290 Agilent Infinity II 2D-LC system from Agilent Technologies, equipped with two 1290 high-pressure binary pumps, a 1290 autosampler with a maximum injection volume of 20 μL, a Multicolumn Thermostat (MCT) column oven and two diode array detectors (DAD) with a 1.0 μL flow cell and a path length of 10 mm. The first and second dimensions were interfaced by a 10-port/4-position ASM valve (configured in backflush mode) from Agilent Technologies, connected via two 1.9 μL transfer capillaries (170 × 0.12 mm) to two 14-port/2-position deck valves carrying either one sample loop with a volume of 180 μL (1870 × 0.35 mm) for the LC-LC experiments, or six sample loops with a volume of 40 μL each (420 × 0.35 mm) for the sLC x LC experiments. The transfer volume, resulting from the time difference between the peak detection in the ^{1}D-DAD and the arrival in the sample loop, which needs to be considered for a correct sampling of the ^{1}D-effluent, was determined experimentally to be 15 μL. The dwell volumes were 180 μL and 90 μL (deck loop volume excluded) in ^{1}D and ^{2}D, respectively.

Experiments were carried out without the pressure release kit (PRK) between the ^{1}D-DAD and the ASM valve, which is normally used to minimize baseline disturbances arising from the 2D-LC valves switching, since the use of this PRK can lead to a reduced ^{2}D-recovery, as discussed in [50]. For experiments without PRK, the ^{1}D-detector was removed as well, in order to protect the flow cell, as recommended by the manufacturer. Stainless steel capillaries of different dimensions from Agilent Technologies were used to obtain different DFs, as shown in **Table S2** in the SI [50]. A JW with a volume of 100 μL from Agilent Technologies was also considered. Agilent OpenLab CDS Chemstation edition (Rev. C.01.10 [287]) software and Agilent 1290 Infinity 2D-LC software add-on (version A.01.04 [036]) were used to operate the system, control the valves and acquire data. Data were analyzed using Agilent 1290 Infinity 2D-LC software and Microsoft Excel.

2.4. General methodology

Separations on both setups were performed with the following mobile phases: (A) 98:2 H_2O:ACN (with AF dissolved in the H_2O-phase and adjusted to pH 3 using FA, to obtain a concentration of 2 mM AF in the H_2O:ACN mixture), (B) 2:98 H_2O:ACN (with AF dissolved in the H_2O-phase and adjusted to pH 3 using FA, to obtain a concentration of 2 mM AF in the H_2O:ACN mixture), (C) 0.1 % FA in H_2O, (D) 0.1 % FA in ACN. All mobile phases were degassed with helium prior to use.

For the ^{1}D-separation, an Acquity BEH HILIC column (1.0 × 150 mm; 1.7 μm) from Waters (Wexford, Ireland) was used in both setups. The ^{1}D-injection volume was 0.5 μL. For the in-house developed setup, the mobile phase was varied as follows: 1/99 – 5/95 – 1/99 – 1/99 (v/v) A/B in 0 – 20 – 20.92 – 30 min at a flow rate of 0.09 mL/min. For the commercial system, an additional isocratic step was added to the gradient program to account for the lower dwell volume of the binary pump. This resulted in the following gradient program: 1/99 – 1/99 – 5/95 – 1/99 – 1/99 (v/v%) A/B in 0 – 3.56 – 23.56 – 24.48 – 30 min. The unretained non-polar compounds on the ^{1}D-HILIC column were sampled and collected in sample loops using switching valves. The different strategies to circumvent the solvent incompatibility of the two dimensions are discussed in Section 2.5.

In the second dimension, the separation was carried out on a Zorbax Eclipse Plus C_{18} column (3.0 × 50 mm; 1.8 μm) from Agilent Technologies in both setups. For the commercial setup, the separation was carried out with the following gradient conditions: 98/2 – 5/95 – 98/2 – 98/2 (v/v%) C/D in 0 - 3.51 - 3.81 - 4.74 min at a flow rate of 0.8 mL/min. For the in-house built setup, a Cortecs Phenyl RP trap column (3.0 × 3 mm; 2.7 μm, Waters) was used in combination with the RPLC column (see Section 2.5). In order to maintain the same gradient slope on both setups, the gradient time for the RPLC separation on the in-house built setup was increased in comparison with the commercial setup. Therefore, the mobile phase was varied using the following gradient conditions: 98/2 – 5/95 – 98/2 – 98/2 (v/v%) C/D in 0 – 4.32 – 4.7 - 8 min at a flow rate of 0.8 mL/min.

The detection wavelength was set at 254 nm in both dimensions with an acquisition rate of 20 Hz. The column oven temperature was not controlled in ^{1}D (room temperature) and was set to 30 °C in ^{2}D of the commercial system.

2.5. Interface: ^{1}D-fraction transfer and dilution strategies

The transfer of the ^{1}D-fractions to the ^{2}D-column and the strategy to overcome the mobile phase incompatibility between HILIC and RPLC differed between both systems, as illustrated in Fig. 1.

2.5.1. In-house developed 2D-LC setup: restriction capillaries

On the in-house developed 2D-LC setup, the sample was only analyzed in LC-LC. **Figure S1** in the SI represents the sequential operation procedure used for this setup. After injection in the ^{1}D (**Figure S1, step 1**), the unretained non-polar compounds were diverted to the sample loop just after eluting from the HILIC column, by switching the configuration of valve 3 (**Figure S1, step 2**). Once the non-polar compounds were collected in the sample loop, valve 3 was switched back to its original position (**Figure S1, step 3**) to direct the separated polar compounds from the HILIC column to the detector. Then, the configuration of the three valves was changed to dilute the solvent containing the non-polar compounds in the sample loop with aqueous mobile phase C (0.1 % FA in H_2O) and direct the non-polar compounds towards the RP trap column (**Figure S1, step 4**). To deal with the solvent incompatibility, the mixing unit developed by Loos et al. was used [26]. This mixing unit consisted of two restriction capillaries with different flow resistances: a high flow resistance capillary (HR, 50 μm x 350 mm) and a low flow resistance capillary (LR, 150 μm x 250 mm), resulting in a DF of ~74. These restriction capillaries were interconnected by two T-pieces creating two parallel flow paths. The different flow resistances of the LR and HR capillaries allowed diluting the ACN-rich ^{1}D-effluent containing the unretained non-polar compounds. To speed up the dilution, a flow rate of 3 mL/min was applied. The dilution allowed trapping the non-polar compounds on the trap column. To ensure all non-polar compounds were transferred to the trap column, the sample loop was flushed at least 1.2 times. After the dilution step, the configuration of valves 1 and 3 was changed to direct the non-polar compounds from the trap column to the RP analytical column in backflush mode (**Figure S1, step 5**).

2.5.2. Commercial 2D-LC setup: active solvent modulation

In contrast to the in-house developed system, the commercial 2D-LC system offers the possibility to carry out LC-LC, sLC x LC, and even LC x LC experiments. For this comparison study, both LC-LC and sLC x LC modes were considered. For LC-LC, operations with no ASM and ASM with different DFs were evaluated. These DFs were obtained using ASM capillaries of different dimensions [50]. **Table S2** in the SI shows the DFs obtained for a sample loop of 180 μL or 40 μL, using different ASM capillaries. For sLC x LC, both no ASM and a DF of 10 were used, as this was the optimal DF in a comparable sLC x LC setup [50]. For the experiments with ASM, the same gradient conditions as for those without ASM were applied, including an additional isocratic step (ASM phase) programmed before the start of the gradient, during which the on-line dilution occurred. For the ASM process, the duration of this isocratic step depended on the DF and on the number of times the sample loop was flushed (λ). An ASM phase duration of $\lambda = 0.5$, and a backflush sample loop unloading configuration were considered optimal for a DF of 10 in a previous study, and were therefore applied in this study as well [50]. For both the LC-LC and sLC x LC mode, ILMM using a JW with a volume of 100 μL from Agilent Technologies (installed between the ASM valve and the ^{2}D-column) was also evaluated with and without ASM. The re-equilibration time was adapted for each experimental setup, to keep a similar equilibration, based on the total ^{2}D-dwell time.

2.6. Calculations

Recoveries were determined by comparing the area under the curve for the peaks in the 2D-LC configuration ($AUC_{2D\text{-}LC}$) with the areas under the curve when the working solution was injected directly into the RPLC column ($AUC_{1D\text{-}RPLC}$) under the same conditions as for the ^{2}D-separation, as shown in Eq. (1):

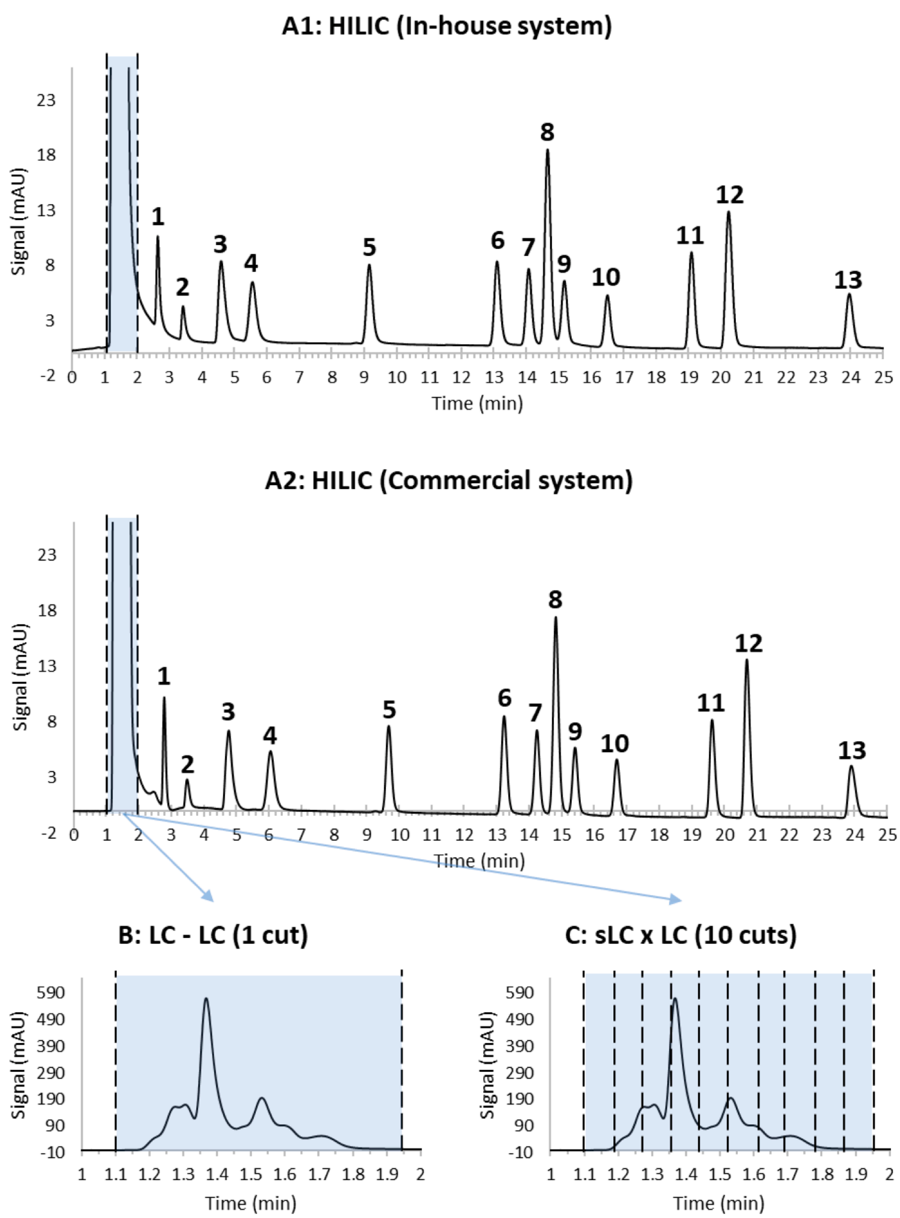

Fig. 2. ^{1}D-HILIC chromatogram and the selected sampling range for transfer to the ^{2}D-RPLC column. A1: Full ^{1}D-HILIC separation on the in-house system. A2: Full ^{1}D-HILIC separation on the commercial system B: Sampling range for heartcutting mode (LC–LC) (same for in-house developed and commercial setup). C: Sampling range for selective comprehensive mode (sLC x LC) on commercial setup.

$$\textbf{Recovery}\ (\%) = \frac{(AUC_{2D-LC})}{(AUC_{1D-RPLC})}\ .\ 100\% \qquad (1)$$

3. Results and discussion

3.1. First dimension: HILIC

To achieve a high degree of comparability, the ^{1}D-HILIC conditions were kept similar between the in-house developed system and the commercial system (an isocratic hold was added to the commercial system to account for the different dwell volumes). In Figs. 2**A1 and A2**, the resulting HILIC separations obtained on both setups are shown. All 13 polar compounds were well-separated, highlighting the benefits of adding a HILIC column to the setup when both very polar and non-polar compounds need to be analyzed in a sample. The non-polar compounds were unretained on the ^{1}D-HILIC column and eluted in a range between 1.1 and 1.95 min on both systems. This unretained fraction was sampled in heart-cutting mode on the in-house developed and commercial setup (Fig. 2**B**), and additionally in selective comprehensive mode on the commercial 2D-LC set-up (Fig. 2**C**).

3.2. Second dimension: RPLC

3.2.1. In-house developed system: LC–LC

One of the strategies to overcome solvent mismatch involves diluting the ^{1}D-effluent with a weak solvent prior to injection on the ^{2}D column. In order to prevent peak distortion and any loss of compounds, it is crucial to thoroughly evaluate the appropriate dilution conditions.

Previous work on the in-house developed setup for a sample of comparable complexity in terms of polarity demonstrated that a DF of 50 was insufficient to adequately dilute and trap the sample plug containing the non-polar compounds on the RPLC trap column [26]. Higher DFs of 74 and 97 resulted in adequate trapping and subsequent satisfactory peak shapes and recoveries of the non-polar compounds on the RPLC column. However, a DF of 97 increased the total analysis time, as the time required to transfer the non-polar compounds from the sample loop to the RPLC trap column increased. Therefore, in this study, a DF of 74 was selected as a trade-off between compound recovery and total analysis time.

As shown in Fig. 2, the non-polar compounds eluted in a range between 1.1 and 1.95 min on the ^{1}D-HILIC column. Considering the flow rate on the HILIC column was 0.09 mL/min, this resulted in a plug volume of 76.5 μL to be transferred to the RPLC column. Due to the parabolic flow profile of the plug in the sample loop, it is recommended that the volume occupied by the collected ^{1}D-fraction should not exceed two-thirds of the loop volume, to avoid losing compounds during transfer [23,51,52]. To investigate the effect of sample loop volume and filling on the obtained recoveries, sample loops with volumes of 100 μL and 150 μL were considered. Additionally, a sample loop volume of 183 μL was also considered, as this was closest to the sample loop volume of the commercial setup (note that the sample loops of the commercial setup cannot be used on the in-house developed setup, due to different fittings). These sample loop volumes hence resulted in sample loop fillings of 77, 51, and 42 %, respectively. Furthermore, to ensure all non-polar compounds were adequately transferred from the sample loop to the trap column, the sample loops were flushed either 1.2 or 1.5 times, cfr. prior optimizations [26]. **Figure S2** in the SI shows the chromatogram obtained by directly injecting the non-polar compounds on the RPLC column, used for the calculation of the recoveries in LC-LC, while the values obtained for the corresponding peak widths can be found in **Table S3**. Fig. 3 shows the separation of the non-polar compounds on the RPLC column (^{2}D) using the in-house developed setup, for a sample loop volume of 183 μL and a flush time of 1.2. **Table S4** in the SI furthermore shows the obtained recoveries and peak widths on the RPLC column in LC-LC, for the different sample loop volumes and flush times.

As can be observed from Table S4, the first eluting compound on the RPLC column (paracetamol, compound 14) was not retrieved under any of the tested conditions. Likely, the employed dilution conditions were insufficient to adequately retain paracetamol on the trap column, hence losing it for subsequent analysis on the RPLC column. Moreover, peak broadening occurred for caffeine (compound 15), ticlopidine (compound 16), and sulfamethoxazole (compound 17), regardless of the sample loop volume and rinsing times applied. Nevertheless, satisfactory recoveries (between 90 and 110 %) were obtained for all compounds across all tested conditions, with the exception of caffeine for the 183 μL sample loop rinsed 1.5 times, however, still resulting in a recovery of 78 %. This is because caffeine, which is relatively polar, is probably (partly) flushed out of the trap column when using a high dilution volume.

Both larger sample loop volumes and longer flush times increased the volume required to dilute the non-polar compounds in the sample loop. As shown in **Table S5,** a higher dilution volume also increased the duration of the dilution step (since the same flow rate was applied for dilution in all cases). However, since the short trap column (with low backpressure) allowed applying a high flow rate of 3.0 mL/min during dilution, this increase in dilution time was relatively modest, increasing

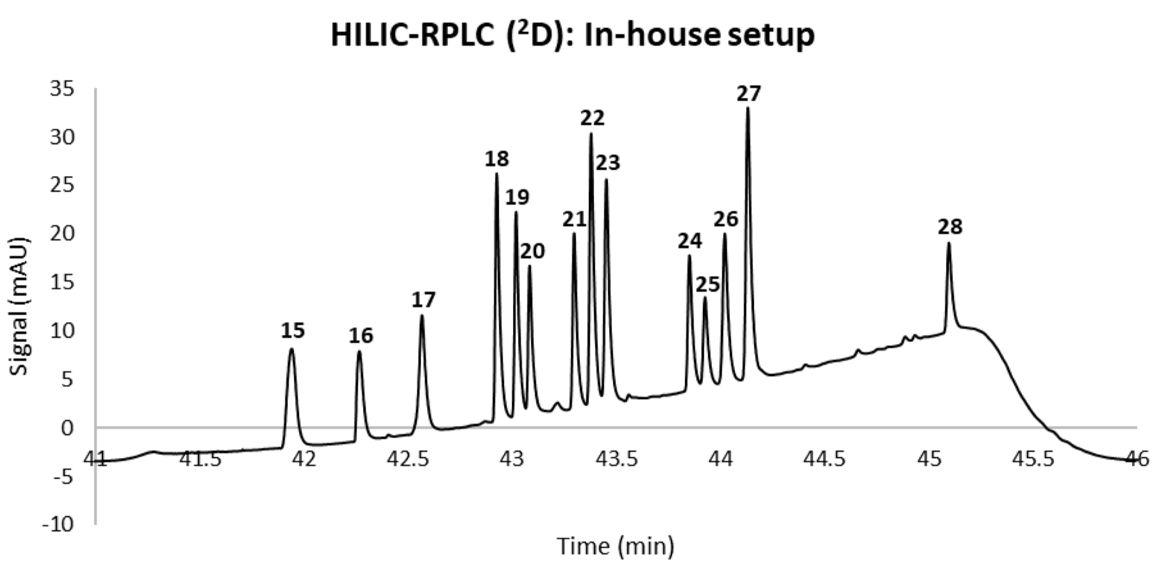

Fig. 3. ^{2}D-chromatogram obtained in HILIC-RPLC setup on the in-house 2D-LC system, using a sample loop volume of 183 μL and a sample loop flush time of 1.2.

the total analysis time from 45 min for a sample loop volume of 100 μL and a flush time of 1.2x, to 49 min for a sample loop volume of 183 μL and a flush time of 1.5x. **Figure S3** in the SI shows the separation of all 28 compounds in a single chromatographic run, achieved by employing a 183 μL sample loop and flush time of 1.2x.

The results demonstrate that the percentage of sample loop filling did not significantly influence the recovery of the non-polar compounds, indicating a high percentage loop filling (77 %) can effectively accommodate the ^{1}D-effluent. A high percentage of sample loop filling moreover allows using relatively small sample loops, reducing the need for excessive dilution volumes and thus overall analysis times. The latter is, however, not too problematic when high flow rates can be applied during dilution. This was the case here, where a trap column with a low backpressure was used in between the sample loop and the analytical column. Overall the in-house developed setup allowed a satisfactory separation of all compounds except one (paracetamol), with acceptable recoveries between 90 and 110 % (in most cases even between 95 and 105 %) in a total analysis time of minimum 45 min, whereby all separated compounds could be visualized in a single chromatogram (for an example see Figure S3).

3.2.2. Commercial system: LC–LC

To compare the commercial 2D-LC setup with the in-house developed setup, the LC-LC approach was considered first because this is essentially the same operational principle as for the in-house developed setup. The main differences with the in-house developed setup are that two pumps are used instead of one (allowing ^{1}D and ^{2}D to be operated simultaneously) and the use of ASM to deal with the mobile phase incompatibility. As for the in-house developed setup, the transferred fraction volume was 76.5 μL, therefore a sample loop filling of 43 % was obtained in the 180 μL sample loop that was available on the setup. The LC-LC approach was carried out without and with ASM, with a sample loop flushing of $\lambda = 0.5$ and a sample loop unloading configuration in backflush mode. Figure S4 in the SI shows the chromatogram obtained by directly injecting the non-polar compounds on the RPLC column, used for the calculation of the recoveries in LC-LC. The corresponding peak widths are given in Table S6 in the SI. In Fig. 4A, the results for the ^{2}D-RPLC separation obtained without ASM are shown. All peaks showed broadening, and severe breakthrough (highlighted by an asterisk) was observed for the earlier eluting compounds (14–23, as specified in Table S1 in the SI) due to the high percentage of ACN in the sample, resulting in poor retention for all compounds.

In Fig. 4B, the results obtained with ASM using a DF of 22, are shown. Other DFs (5 – 64) were tested as well, as shown in Figure S5 in the SI. In Table S7 in the SI, all values regarding the obtained recoveries and peak widths are shown. A DF of 22 gave the best peak shapes while keeping the analysis time minimal, as illustrated in Fig. 4B. Note that this DF is lower than the optimal DF used in the in-house developed setup, since the non-polar compounds were immediately sent to the analytical column in the commercial setup, which had a higher retention capacity

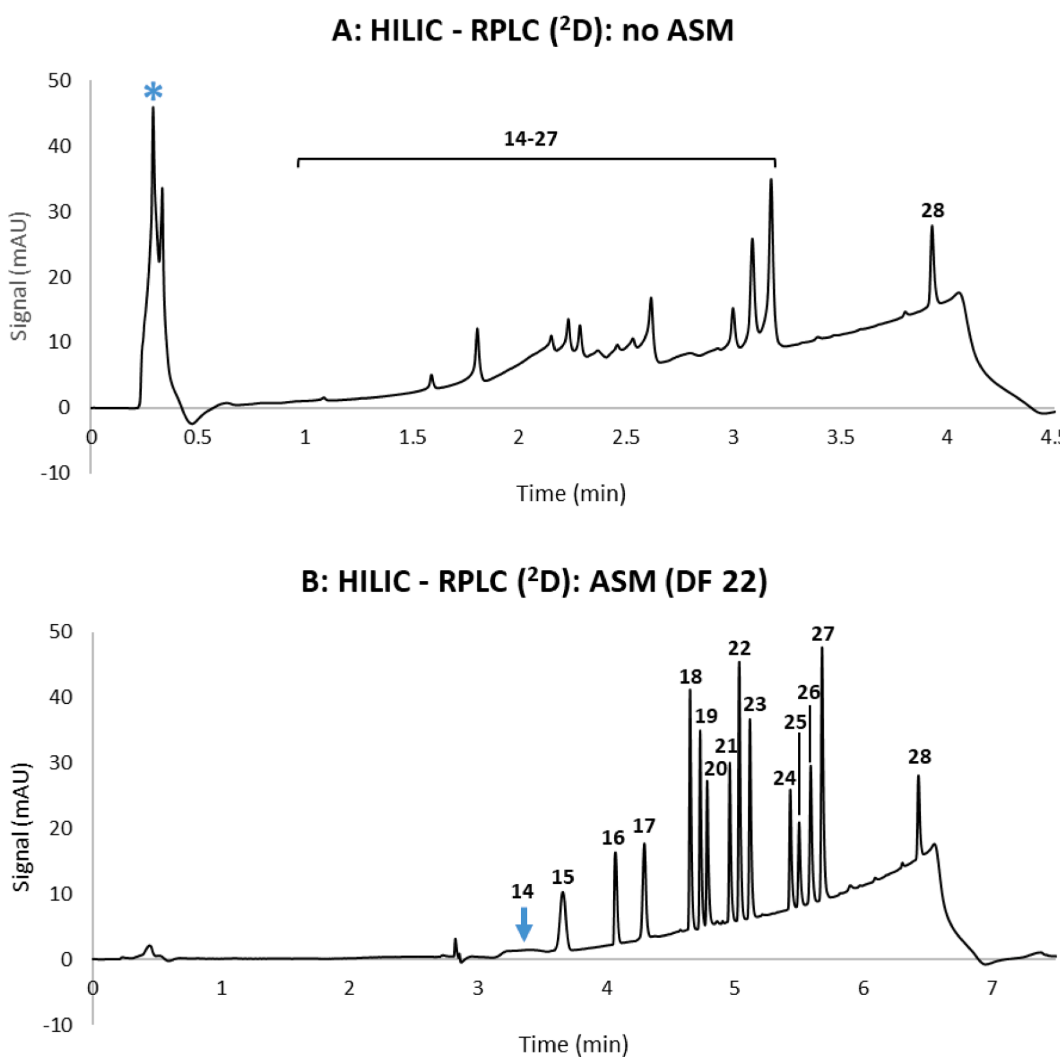

Fig. 4. Comparison of ^{2}D-chromatograms obtained with/without ASM in HILIC-RPLC setup on commercial 2D-LC system. A: no ASM. B: ASM (DF 22).

compared to the trap column in the in-house developed setup. The later eluting compounds (18–28) displayed sharper peak shapes compared to the same conditions without ASM. However, earlier eluting compounds (15–17) still showed some peak broadening, while the first eluting compound (compound 14, paracetamol) was missing from the chromatogram (indicated by an arrow in Fig. 4B), very similar to what was observed for the in-house developed setup. Recoveries were slightly worse compared to the in-house developed setup and ranged between 85 and 113 % for a DF of 22. Diluting the ^{1}D-fraction with the starting gradient composition of the ^{2}D-mobile phase (2:98 ACN:H_2O) during the ASM phase, hence significantly improved the peak shapes, especially for the later eluting compounds, due to a better focusing effect at the column inlet upon injection. These improved peak shapes, however, came at the cost of an increased ^{2}D-analysis time, due to the extra time required for on-line dilution (0.6 min for a DF of 5, 1.22 min for a DF of 11, 2.5 min for a DF of 22 and 7 min for a DF of 64), resulting in a total analysis time of 5.52 min for a DF of 5, 6.14 min for a DF of 11, 7.42 min for a DF of 22 and 12.12 min for a DF of 64. For the early eluting compounds, the resulting mobile phase composition after dilution remained too strong and the volume of the fraction too large, therefore still causing poor peak shapes, even after dilution. Especially for the first compound, for which the sample plug was so large and the resulting analyte band too broad to result in a visible peak in the ^{2}D-chromatogram.

A combination of ILMM and ASM was therefore assessed next, as shown in Figure S6 and Table S8 in the SI. The peak shapes obtained when using a JW without ASM already improved compared to the results obtained without ASM (Figure S6A versus Fig. 4A). However, the combination of a JW and ASM was not better compared to using solely ASM (Figure S6B versus Fig. 4B), most likely due to the large volume of the ^{1}D-fraction (76.5 μL) compared to the volume of the JW (100 μL), insufficient to create adequate mixing and dilution. Overall, these results demonstrate that diluting the mobile phase of the transferred fraction before directing it to the second dimension is a good approach to tackle the mobile phase incompatibility between HILIC and RPLC, as was also demonstrated with the in-house set-up. A great advantage of the commercial system is that the ^{1}D- and ^{2}D-separations are carried out almost simultaneously, since the ^{2}D-separation can immediately start once the ^{1}D-fraction has been collected, resulting in a shorter total analysis time: 30 min for the commercial setup (time of the ^{1}D-HILIC

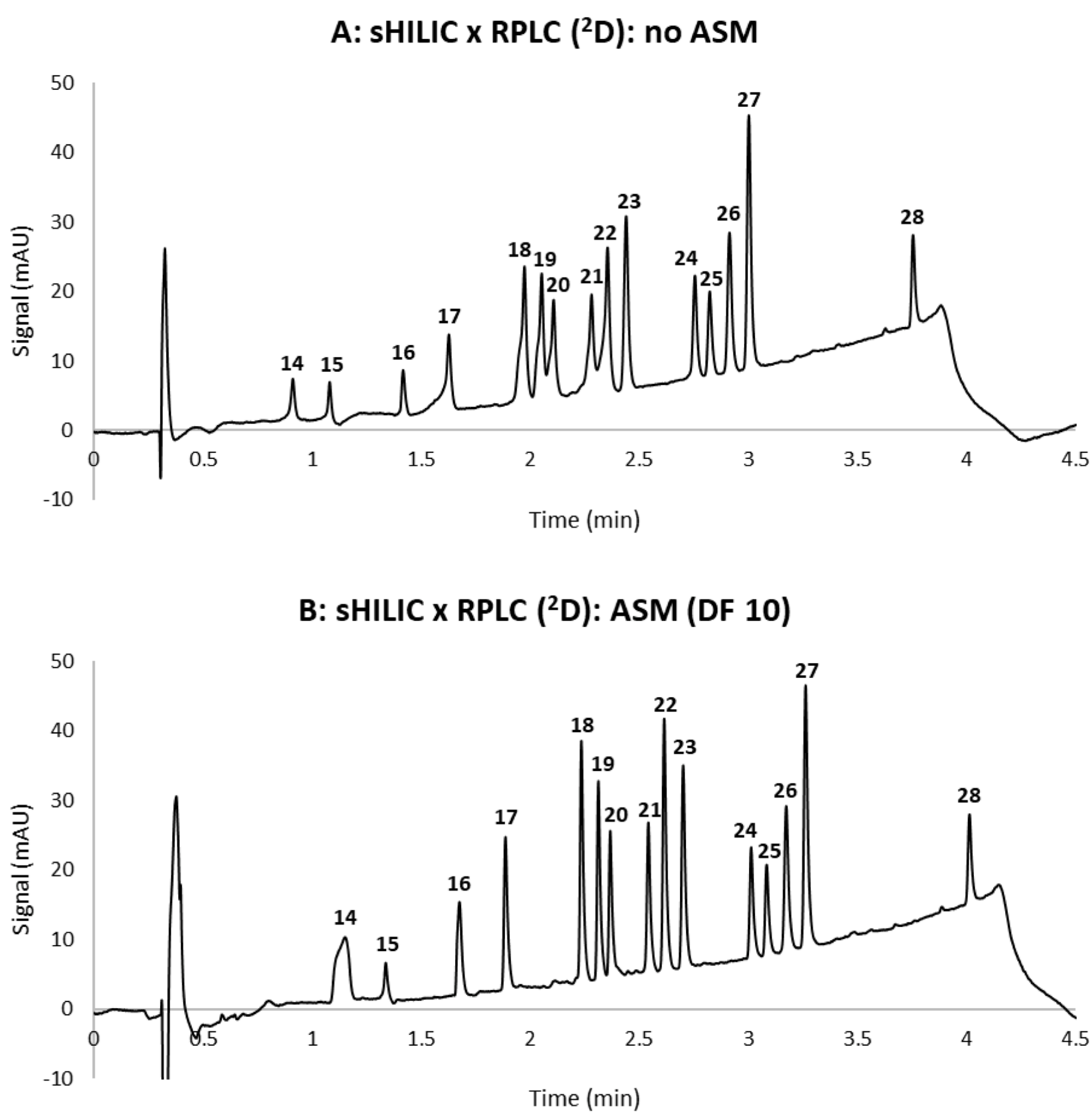

Fig. 5. Comparison of ^{2}D-chromatograms obtained with/without ASM in sHILIC x RPLC setup on commercial 2D-LC system. A: no ASM. B: ASM (DF 10). Both chromatograms show the sum of the cuts taken from the first dimension for each condition.

separation) versus 45 min for the in-house setup (^{1}D-HILIC separation followed by trapping and ^{2}D-RPLC separation). On the other hand, for the in-house setup, the entire separation is shown in a sequential manner in the same chromatogram, which facilitates the integration and interpretation of the results. Overall, the results obtained on both setups are still not optimal, mainly for the earlier eluting compounds that suffered more from the incompatibility issue. Therefore, it was investigated whether this could be improved by carrying out an sLC x LC experiment.

3.2.3. Commercial system: sLC x LC

Due to the presence of the two decks containing multiple sample loops, the commercial system can also be used for sLC x LC experiments, in addition to the other 2D-LC modes.

The sLC x LC cuts resulted in 10 transferred fractions of 7.65 μL each, therefore a sample loop filling of 19 % was obtained for the 40 μL sample loops. Reducing the volume of the transferred ^{1}D-fractions is a possible strategy to further reduce the negative effects on the ^{2}D-peak shapes when dealing with mobile phase incompatibility problems [23]. In Fig. 5**A**, the results for the sLC x LC approach obtained without ASM are shown, with a sample loop flushing of $\lambda = 0.5$ and a sample loop unloading configuration in backflush mode. In Fig. 5**B**, the results obtained with ASM for a DF of 10 are shown. This DF was previously shown to be ideal in an sLC x LC setup under similar conditions [50]. In **Table S9** in the SI, all values regarding the recoveries and peak widths can be found. It is clear that, even without on-line dilution, sLC x LC gave better peak shapes compared to the LC-LC approach without ASM. This is due to the reduced volume of the transferred ^{1}D-fractions (10 transferred fractions of 7.65 μL each, compared to one transferred fraction of 76.5 μL in the LC-LC mode). Additionally, the filling percentage of the sample loops was also lower, resulting in additional in-loop dilution of the sample plug with the ^{2}D-mobile phase, improving the ^{2}D-peak shapes further [50].

With ASM, the results were even better, and the first compound (14) was now better focused on the ^{2}D-column and therefore visible in the chromatogram, as opposed to the LC-LC method. Despite these improved peak shapes, the total sLC x LC analysis time (56 min) was much longer than that of the commercial LC-LC method (30 min) and the LC-LC method on the in-house developed setup (45 min). The reason is that the total sLC x LC analysis time is the sum of the analysis time of each cut (5 min) and the time needed to flush the capillaries in the decks (which is always programmed into the analysis by the software, and is equal to the time of one ^{2}D-cycle). Overall, it seems that a tradeoff needs to be made between minimizing analysis time and achieving the best peak shapes for early eluting compounds.

Even though the first compound was now focused and visible on the ^{2}D-column, the peak shapes and recoveries of the first two eluting compounds were still not optimal. For caffeine, the recovery was even substantially lower than for the in-house and commercial LC-LC methods. Several possibilities could be considered to further improve these [50], including changing the dilution solvent to 100 % H_2O (using a different stationary phase which can operate at 100 % H_2O in the second dimension). However, in this study, the dilution solvent was already 98 % H_2O, therefore the additional benefit would probably be minimal. Another option would be to increase the retention factor of the early eluting compounds by selecting a different stationary phase, however, at the risk of changing the separation selectivity. ILMM using the JW either with or without ASM was assessed as well, as shown in **Figure S7** in the SI. Both approaches resulted in somewhat better peak shapes compared to the optimized ASM conditions for the first two eluting compounds, while their ^{2}D-recoveries were significantly improved (slightly more for the JW in combination with ASM), as shown in **Table S10** in the SI. It has been shown before that the performance of ILMM can be better than ASM alone, depending on the volume of the transferred fraction and that of the JW used, since the multichannel flow path enables a more effective homogenization of ^{1}D-fractions, thus allowing the analytes to be more focused [39]. An improvement was only observed when adding the JW in the sLC x LC mode compared to the LC – LC mode, since much smaller fraction volumes were used in sLC x LC, while the fraction volume in LC-LC was rather of the same order of magnitude as that of the JW. Overall, the volume of the JW versus that of the transferred fractions is an important parameter in the method optimization, and increasing the JW volume for LC-LC setups with larger fractions might improve the ^{2}D-peak shapes even more. Additionally, the impact of the JW on the total analysis time was limited: using no ASM and the JW required 4.87 min per cut, resulting in a total analysis time of 55 min, only ASM (DF 10) required 5 min per cut, resulting in a total analysis time of 56 min, while the combination of ASM and JW required 5.1 min per cut, resulting in a total analysis time of 57.5 min.

3.3. General differences between the in-house developed and the commercial 2D-LC system

In addition to the experimentally determined differences in peak shapes and analysis time, some other general facts need to be considered in this comparison. One of the most important differences is the cost price of both systems: the commercial system is significantly more expensive than the in-house developed setup (2023 list price for commercial setup (excl. VAT): 270.000 Euro (2 binary high-pressure pumps, 2 DAD, 1 vialsampler, 2D-LC interface and software) versus 140.000 Euro for the in-house developed setup (1 quaternary high-pressure pump, 1 DAD, 1 vialsampler, 2D-LC interface and software). This is mainly due to the need for two pumps and two detectors, while the in-house system uses a (cheaper) quaternary pump to supply the mobile phases for both dimensions. Both separation dimensions are carried out sequentially in the in-house developed setup, while both separation dimensions are carried out simultaneously in the commercial setup, necessitating the use of two separate pumps. The latter approach has the advantage that the second dimension separation can already be initiated while the first dimension separation is still ongoing, which saves time and makes LC x LC (and sLC x LC) possible. These 2D-LC modes, however, generate a more complex output (multiple ^{2}D-chromatograms, which need to be summed up in the case of sLC x LC or aligned to generate a contour plot in LC x LC). Such software needs to be purchased separately (additional price of 2D-LC add-on and GC Image LCxLC software: 18.000 Euro), further increasing the cost of the commercial system. For the 2D-LC interface, both setups require 3 additional valves. For the in-house developed setup, this results in an additional cost of around 28.000 Euro (3 valve drives and valve heads), while this price is significantly larger for the commercial system (51.000 Euro), due to the cost of the 2D-LC ASM valve kit and the two multiple heart-cutting valves.

Additionally, since the commercial system can simultaneously perform the ^{1}D- and ^{2}D-separations, this implies that two detectors are required for LC-LC and sLC x LC experiments, which is not problematic if DADs are used (except in terms of an increased cost price). If a destructive detector such as a mass spectrometer, evaporative light scattering detector (ELSD) or charged aerosol detector (CAD) is used, the experiments need to be carried out twice if information about both dimensions is required, whereby in each run a different dimension is coupled to the detector. This will increase the total analysis time significantly. When performing LC x LC experiments, this is not an issue, since the entire first dimension is also sent to the second, so only the ^{2}D-column needs to be connected to a detector.

Overall, the possibility of performing LC x LC experiments, which is usually needed for the analysis of very complex samples, is a main advantage of the commercial system. An LC x LC setup can also be lab-made, however, this requires two pumps and is most likely not as robust and user-friendly as the commercial options. This implies that the commercial system can easily be used for many different types of samples with different degrees of complexity, depending on the analytical goal (LC-LC, sLC x LC or LC x LC).

In terms of instrument operation, an in-house 2D-LC system can be built up from an existing 1D-LC system by adding the required switching valves, capillaries, columns, and sample loops, thus providing adaptability to specific analytical goals, and making it interesting for more innovative or complicated 2D-LC research. This, however, requires sufficient knowledge of LC systems and can therefore be less straightforward for beginning practitioners. On the other hand, the commercial system is bought as a whole, which makes it less inviting to make changes to the standard setup, although the latter is still possible (e.g., introducing a JW for ILMM or further customizing the ASM capillaries). Generally, the commercial system is easier for less experienced users, and more routine applications. In addition, the specialized 2D-LC software developed for these systems makes setting up 2D-LC experiments more straightforward.

In comparison, when using an in-house setup, there is no need for (expensive) specialized software, but it is necessary to manually configure the method, which is more time-consuming, and carefully monitor the chromatographic run.

When dealing with the mobile phase incompatibility, the commercial system has some advantages. Since multiple sample loops are available, smaller transferred fractions and required DFs are possible, which overall results in better ^{2}D-peak shapes. It is perfectly possible to add additional sample loops to the in-house developed system as well, to accommodate multiple smaller fraction volumes. However, this would require the introduction of two more switching valves, thus increasing the price, bulkiness, and complexity of the system.

Finally, the commercial systems are very robust and provide highly reproducible 2D-LC experiments, while the in-house system is more fragile and technical problems can occur more easily when the setup is less well understood. In the commercial system, the delay time is also significantly reduced.

4. Conclusion

The goal of this study was to make a comparison between an in-house developed 2D-LC setup and a commercially available 2D-LC instrument, based on experimental differences, in addition to more general differences, including cost price, flexibility, and ease of operation. The different strategies to deal with the mobile phase incompatibility between HILIC and RPLC was investigated as well in detail on both setups. For the commercial 2D-LC instrument, the strategy used was ASM and ILMM, while for the in-house developed setup, a combination of restriction capillaries and a trap column was used. Using a sample of 28 compounds with a large polarity range, peak shapes and recoveries of the ^{2}D-chromatograms were compared for both setups, as these are important parameters for an adequate quantification of compounds of interest in real applications. For early eluting and hence very polar compounds, the sLC x LC approach, currently only possible on the commercial 2D-LC instrument, resulted in the best peak shapes and recoveries, however, at the cost of an increased total analysis time of 56 min. The ILMM-approach performed slightly better compared to ASM, as demonstrated before [39]. For later eluting compounds, there were no significant differences between the instruments in terms of peak shapes and recoveries. In this case, the most advantageous setup in terms of analysis time was the LC - LC approach on the commercial instrument (total analysis time of 30 min versus 45 min on the in-house developed system), since the ^{1}D- and ^{2}D-separation could be carried out simultaneously. On the other hand, in terms of total cost, the in-house developed setup was the least expensive, with a list price that was almost half that of the commercial setup. In general, depending on the analytical goal (single heart-cut versus full-comprehensive 2D-LC), an in-house developed system can be satisfactory for the analysis of specific target compounds/samples. For more complex problems and flexibility, it can be interesting to purchase a more specialized commercial 2D-LC instrument. Finally, this study focused on the comparison of two systems from the same vendor, as these were available in our lab and made the comparison in terms of cost price more straightforward. Systems from other vendors are of course also available, and could provide additional capabilities in terms of column switching and strategies to modify the composition of the fractions transferred between both dimensions. Investigating and comparing the possibilities of such systems could provide directions for future research. Additionally, this study focused on a very particular sample, representative for typical environmental analyses. Other sample types, containing an even wider variety of compound classes, could shed further light on the capabilities of in-house versus commercially available systems. The main purpose of this study was to encourage analysts to explore the capabilities of the systems available in their labs and select the most appropriate set-up (in-house built versus commercially available), depending on the particular needs of their samples, their own experience and skills, and the available budget for instrumentation.

CRediT authorship contribution statement

Marie Pardon: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft. **Rafael Reis:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft. **Peter de Witte:** Funding acquisition, Supervision, Writing – review & editing. **Soraya Chapel:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Supervision, Writing – review & editing. **Deirdre Cabooter:** Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

M.P. is funded by the Research Foundation Flanders (FWO) (SBO research project “SmartDetox”, project ID: 3E200795). R.R. is funded by the European Union’s EU Framework Program for Research and Innovation Horizon 2020 [Grant Agreement No 861369, 2019, innoveox.eu]. S.C. is funded by the joint-initiative of the Research Foundation Flanders (FWO) and the Walloon Fund for Scientific Research (FNRS) (EOS – research project “Chimic” (EOS ID: 30897864)).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.chroma.2023.464565.

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